

Car Sales Analysis Dashboard Using Microsoft Excel

Abstract

This project aims to analyze car sales data and develop an interactive dashboard using Microsoft Excel. The dashboard provides insights into sales performance, profitability, dealer efficiency, and customer preferences. Through data cleaning, pivot tables, and visualizations, the project demonstrates how Excel can be used as an effective business intelligence tool to support strategic decision-making.

Introduction

In today's competitive automobile industry, data-driven decision-making is essential for improving sales performance and profitability. Organizations generate large amounts of sales data, which can be transformed into valuable insights through data analysis and visualization techniques. This project focuses on creating a Car Sales Analysis Dashboard to evaluate business performance and identify growth opportunities.

Problem Statement

Businesses often struggle to analyze large volumes of sales data and identify factors affecting profitability and sales performance. There is a need for an interactive dashboard that can provide quick and meaningful insights into sales trends and business performance.

Objectives

- Analyze overall sales performance.
- Identify top-performing car models.
- Evaluate dealer performance.
- Study monthly and yearly sales trends.
- Compare sales volume and profitability.
- Develop an interactive dashboard for business decision-making.

Dataset Description

The dataset contains transaction-level information, including:

- Date
- Year and Month
- Car Model
- Dealer ID
- Quantity Sold

- Profit

Year	Month	Date	Model	Dealer ID	Quantity Sold	Profit
2018	Feburary	2018-02-01 23:00:00	Labrador	1215	22	17600
2018	Feburary	2018-02-01 23:00:00	Champlain	1215	4	9200
2018	Feburary	2018-02-01 23:00:00	Hudson	1215	32	16000
2018	Feburary	2018-02-01 23:00:00	Beaufort	1301	15	22500
2018	Feburary	2018-02-01 23:00:00	Salish	1301	20	61000
2018	Feburary	2018-02-01 23:00:00	Labrador	1301	26	20800
2018	Feburary	2018-02-01 23:00:00	Champlain	1301	5	11500
2018	Feburary	2018-02-01 23:00:00	Hudson	1301	36	18000
2018	Feburary	2018-02-01 23:00:00	Beaufort	1336	13	19500
2018	Feburary	2018-02-01 23:00:00	Salish	1336	17	51850
2018	Feburary	2018-02-01 23:00:00	Labrador	1336	21	16800
2018	Feburary	2018-02-01 23:00:00	Champlain	1336	4	9200
2018	Feburary	2018-02-01 23:00:00	Hudson	1336	30	15000
2018	Feburary	2018-02-01 23:00:00	Beaufort	1401	12	18000
2018	Feburary	2018-02-01 23:00:00	Salish	1401	16	48800

Methodology

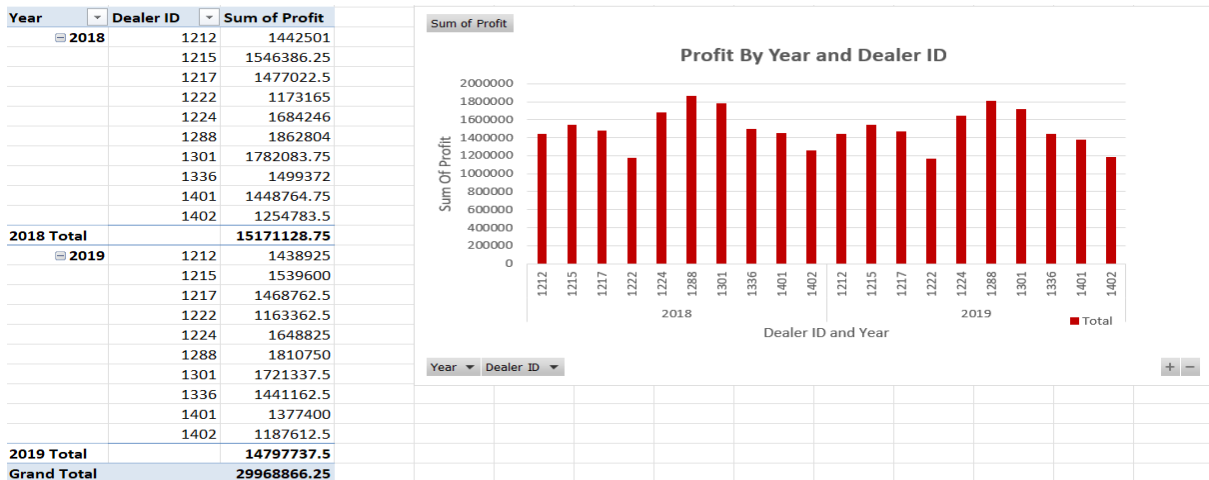
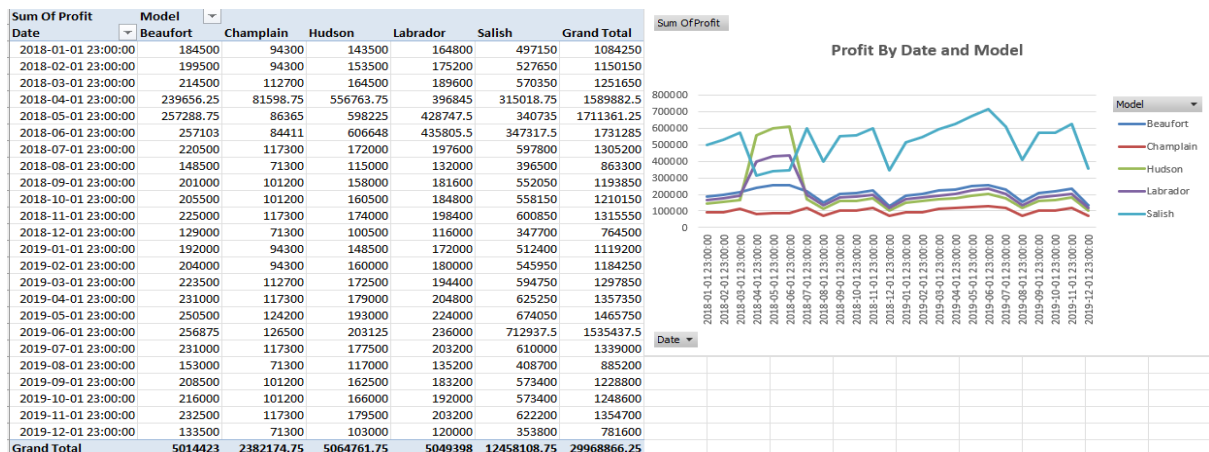
1. Data Collection
2. Data Cleaning and Transformation
3. Creating Pivot Tables
4. Building Visualizations
5. Dashboard Development
6. Generating Business Insights

Tools Used

- Microsoft Excel
- Pivot Tables
- Pivot Charts
- Data Visualization Techniques

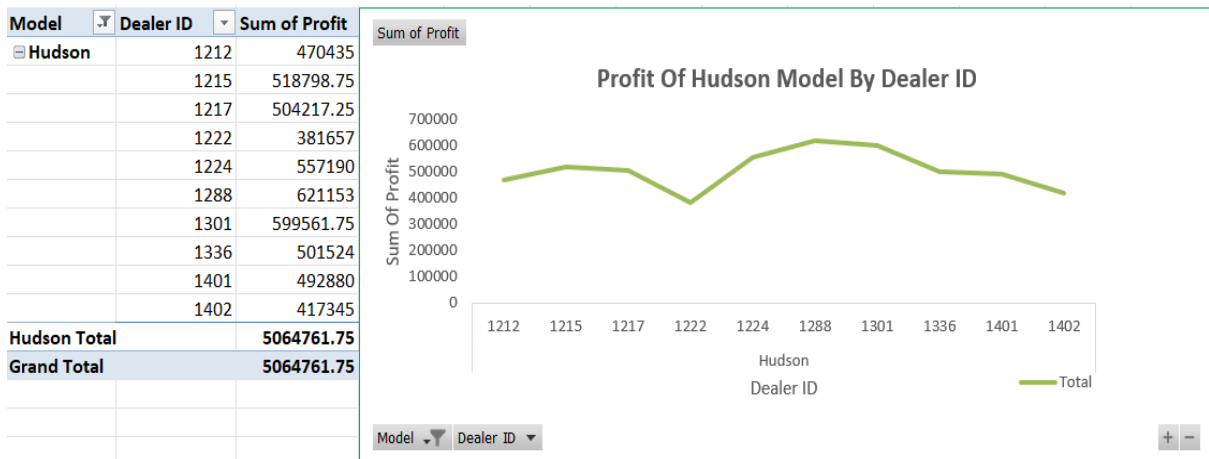
Analysis and Findings





- Total Units Sold: **21,597**
- Total Profit Generated: **₹2,99,68,866+**
- Analysis covers **2 years of data and 5 car models.**

Top Performing Model

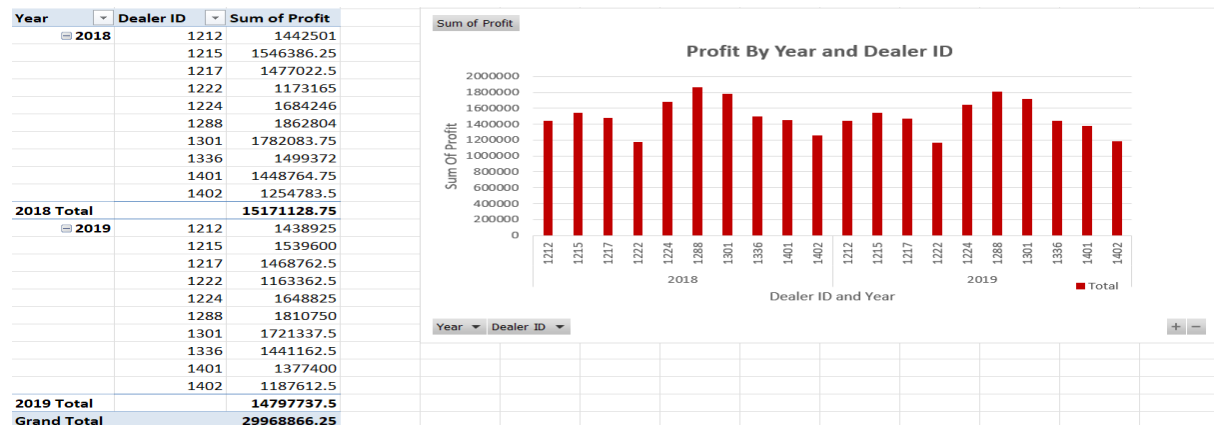


Hudson was identified as the best-selling car model, indicating strong customer demand.

Sales Trends

Monthly fluctuations reveal seasonal demand patterns and potential peak sales periods.

Dealer Performance



Significant differences exist among dealer performances, highlighting opportunities for performance improvement and strategy replication.

Profit Analysis

Higher sales volume does not necessarily result in higher profitability, emphasizing the importance of focusing on profit margins.

Business Recommendations

- Increase inventory for high-demand models.
- Replicate strategies of top-performing dealers.
- Launch marketing campaigns during peak seasons.
- Focus on high-margin products to improve profitability.

Conclusion

The project successfully demonstrates how Microsoft Excel can be used to transform raw sales data into actionable business insights. The interactive dashboard enables organizations to monitor sales performance, identify trends, and make informed decisions that enhance profitability and operational efficiency.

Future Scope

- Integration with Power BI for advanced analytics.
- Sales forecasting using predictive models.
- Real-time dashboard updates through database connectivity.